

DevNexus | AgileBridge

UPDATED USE CASES

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agilebridge
CONSIDERED
SOFTWARE SOLUTIONS

3. Use Case Diagrams

UC1: User and Identity Management

UC1.1: Register Account

- **TUCBW** a prospective student, not registered in the system yet and has a valid South African university email address, submits the registration form with all the required information.
- **TUCEW** the user account is created, but the user is placed in a “Pending verification” state (unverified). An OTP is sent to the provided student email.

UC1.2: Get OTP

- **TUCBW** the user has successfully registered and is on the OTP verification page.
- **TUCEW** a One-Time Password (OTP) is generated and sent to the user’s registered email address.

UC1.3: Verify OTP

- **TUCBW** the user has received the OTP via email and has pressed verify on the verification screen.
- **TUCEW** the entered OTP is validated. The user moves to the next step of the verification workflow (Uploading Proof of Registration).

UC1.4: Upload Proof of Registration

- **TUCBW** the user successfully verified their email via OTP, and the user is on the upload Proof of Registration screen.
- **TUCEW** the Proof of Registration file is uploaded to the system, and the verification process is initiated.

UC1.5: AI Verify Proof

- **TUCBW** the user has successfully uploaded their Proof of Registration document.
- **TUCEW** the AI processes the Proof of Registration and outputs a confidence score, which determines the next action.

UC1.6: Manual Verification

- **TUCBW** a user’s account or listing has been added to the admin queue (due to Medium/Low AI confidence, Listing Risk, or Manual Report).

- **TUCEW** The admin makes a decision (Approve/Reject/Request Changes), updating the user's status or listing visibility.

UC1.7: Resubmit Proof of Registration

- **TUCBW** the user has been flagged for manual review or rejected, and the admin has requested a new/updated proof of registration.
- **TUCEW** the new file is uploaded to the system; the user status is reset to "Pending verification".

UC1.8: Login

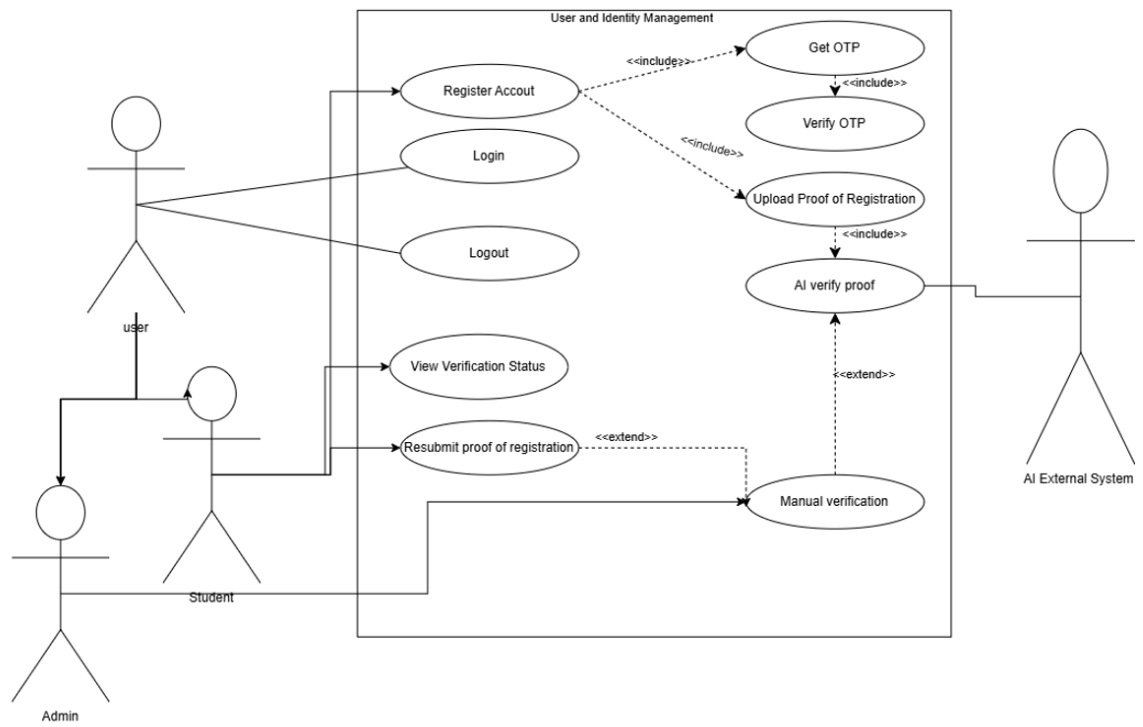
- **TUCBW** a registered student enters their valid credentials (email and password) and press Login.
- **TUCEW** the user is granted access to the system. Access permissions are determined by their verification status (e.g., Partial Access if unverified, Full Access if verified).

UC1.9: View Verification Status

- **TUCBW** a logged in user navigates to their profile page.
- **TUCEW** the user sees their current verification status ("Pending", "Verified", "Rejected" or "Review Required").

UC1.10: Logout

- **TUCBW** the user who is currently logged into an active session, presses Logout from their profile page or the sidebar.
- **TUCEW** the user's session is terminated, and the user is redirected to the Login screen.



UC2: Listing and Inventory Management

UC2.1: Upload a Listing

- **TUCBW** the seller is logged in, and they are on the upload listing page, then they fill in the listing details (title, description, price, condition, course tags), upload photos and press Submit.
- **TUCEW** the listing is saved as draft (invisible to buyers), if the seller is verified, the system automatically proceeds to the AI risk-scoring pipeline. If the seller is unverified, the listing remains as a draft until they are verified.

UC2.2: Edit Active Listing

- **TUCBW** the seller has an existing listing that is currently live (visible to buyers), and the seller presses Edit Listing in the Listings' table.
- **TUCEW** the changes are saved and reflecting in the listings' table.

UC2.3: Save Listing Draft

- **TUCBW** the seller has entered only the information available to them and pressed save as draft.
- **TUCEW** the partially filled listing data persisted as a draft in the system, and the listing is reflecting on the listings' table with a draft status.

UC2.4: Remove Listing

- **TUCBW** the seller has pressed the delete icon on an of the listing in their listings' table.
- **TUCEW** the listing is permanently deleted from the platform, and it is removed from the buyer's browse listing page (if it was live).

UC2.5: View all Listings

- **TUCBW** the seller is logged into the system and navigates to the My Listings Page.
- **TUCEW** the system displays all the seller's uploaded listings.

UC2.6: Acknowledge Reservation

- **TUCBW** the buyer has reserved the item, and the seller sees the reservation on their Reserved page and pressed acknowledge.
- **TUCEW** the chat thread has opened, and the buyer and seller can chat back and forth.

UC2.7: Cancel Reservation

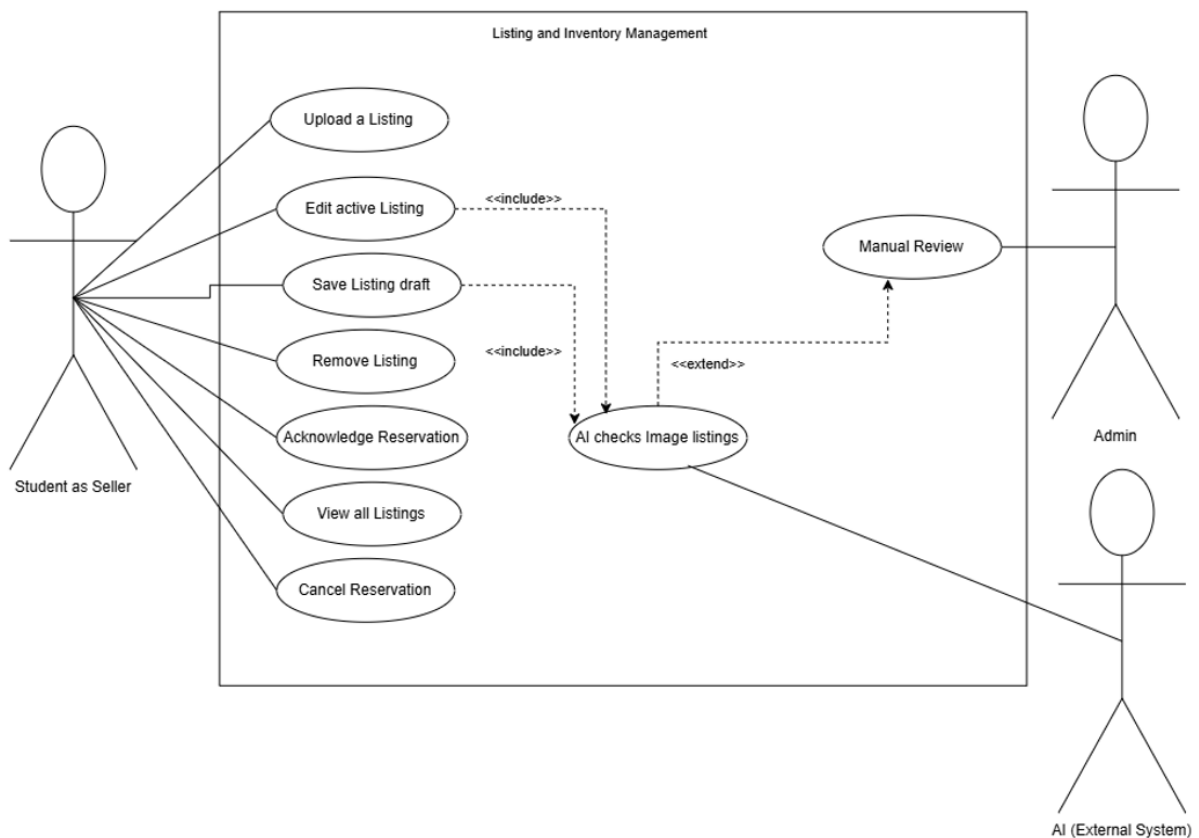
- **TUCBW** the seller cancels an active reservation after 12 hours of buyer silence.
- **TUCEW** the reservation is voided. The listing is immediately released back to the marketplace, making it available for other buyers to reserve.

UC2.8: AI checks Image listing

- **TUCBW** the seller has uploaded a listing.
- **TUCEW** the AI processes the listing photos and outputs a confidence score, which determines the next action.

UC2.9: Manual Review

- **TUCBW** the seller's listing is in the Admin Listing queue (due to Medium/Low AI confidence)
- **TUCEW** the admin makes a decision (Approve/Reject/Request Changes), updating the listing's status and visibility.



UC3: Discovery and Recommendations

UC3.1: View dashboard

- **TUCBW** the student (Guest or Verified) is logged in and navigates to the main marketplace feed.
- **TUCEW** the dashboard loads, displaying all published/live listings.

UC3.2: Generate suggested materials

- **TUCBW** the student is logged in and on the Dashboard.
- **TUCEW** the system displays a personalized list of recommended listings (e.g., “Popular for COS151 at University X” or “Frequently bought together”).

UC3.3: Search Listing

- **TUCBW** the student enters a keyword (title) into the search bar on the marketplace.
- **TUCEW** the system returns a list of live listings matching the query.

UC3.4: Sort Results

- **TUCBW** the student has performed a Search or is viewing a category and selects a sorting option.
- **TUCEW** the displayed list of listings is re-ordered according to the selected criteria.

UC3.5: Filter Results

- **TUCBW** the student is viewing the marketplace or search results and applies filter criteria.
- **TUCEW** the system narrows down the listing results to show only the items matching the selected filters.

UC3.6: View Material Details

- **TUCBW** the student clicks on a specific listing card from the Browse All Listings page, search results, or Wishlist.
- **TUCEW** the system shows the listing’s details.

UC3.7: Add to Wishlist

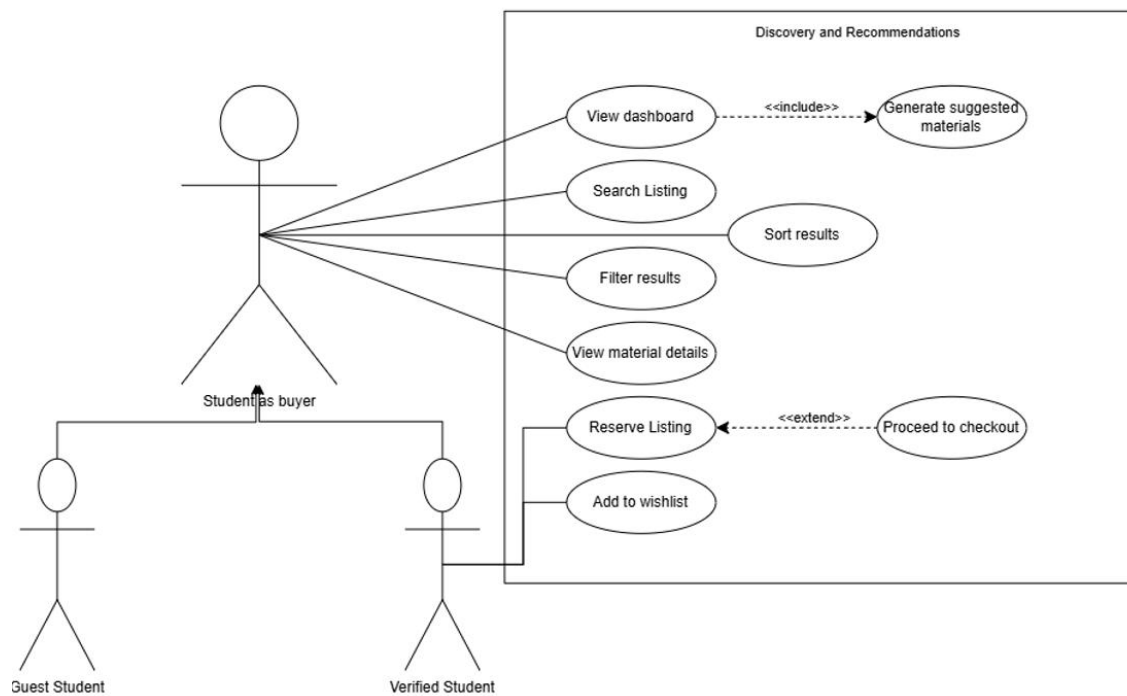
- **TUCBW** a logged in user is viewing the details of a live listing, and presses “Add to Wishlist”.
- **TUCEW** the listing is saved to the student’s personal Wishlist for future quick access.

UC3.8: Reserve Listing

- **TUCBW** the verified student presses “Reserve” on the available listing’s card.
- **TUCEW** the system creates a temporary reservation, the 24-hour timer starts, and an automated initial “interested” message is sent to the seller via the chat.

UC3.9: Proceed to Checkout

- **TUCBW** a verified student with an active reservation, has coordinated a meetup with the seller clicks “Pay” on the Meetup Details Page.
- **TUCEW** the buyer is redirected to the Payfast payment gateway to complete their payment.



UC4: Reservation and Transaction

UC4.1: Reserve Item

- **TUCBW** the verified student/buyer views a listing and selects “Reserve”
- **TUCEW** an automated initial “interested” message is generated and sent to the seller, opening a chat thread and placing a 24-hour initial reservation timer on the item.

UC4.2: Cancel Reservation

- **TUCBW** the buyer manually chooses to cancel their active reservation, presses “Cancel” on the reservation card.
- **TUCEW** the reservation is removed, and the item becomes available to other buyers again.

UC4.3: Reserve full degree list

- **TUCBW** the buyer chooses to reserve a bundle/full degree list of textbooks from a single seller.
- **TUCEW** a 7-day reservation window is initiated for the bundle, and the seller is notified.

U4.4: Purchase item

- **TUCBW** the buyer attends the meetup, inspects the physical item, approves its condition, and initiates payment.
- **TUCEW** the payment is redirected to the payment gateway (PayFast) to process.

UC4.5: Process payment

- **TUCBW** the payment flow is initiated via PayFast.
- **TUCEW** the PayFast backend sends a successful webhook response to the system, verifying the payment and generating a confirmation PIN for the buyer.

UC4.6: Confirm Receipt

- **TUCBW** the seller shows the generated PIN to the buyer at the meetup.
- **TUCEW** the buyer enters the PIN into the app, closing and completing the transaction.

UC4.7: Raise Dispute

- **TUCBW** a buyer identifies a problem and submits a report/flag.

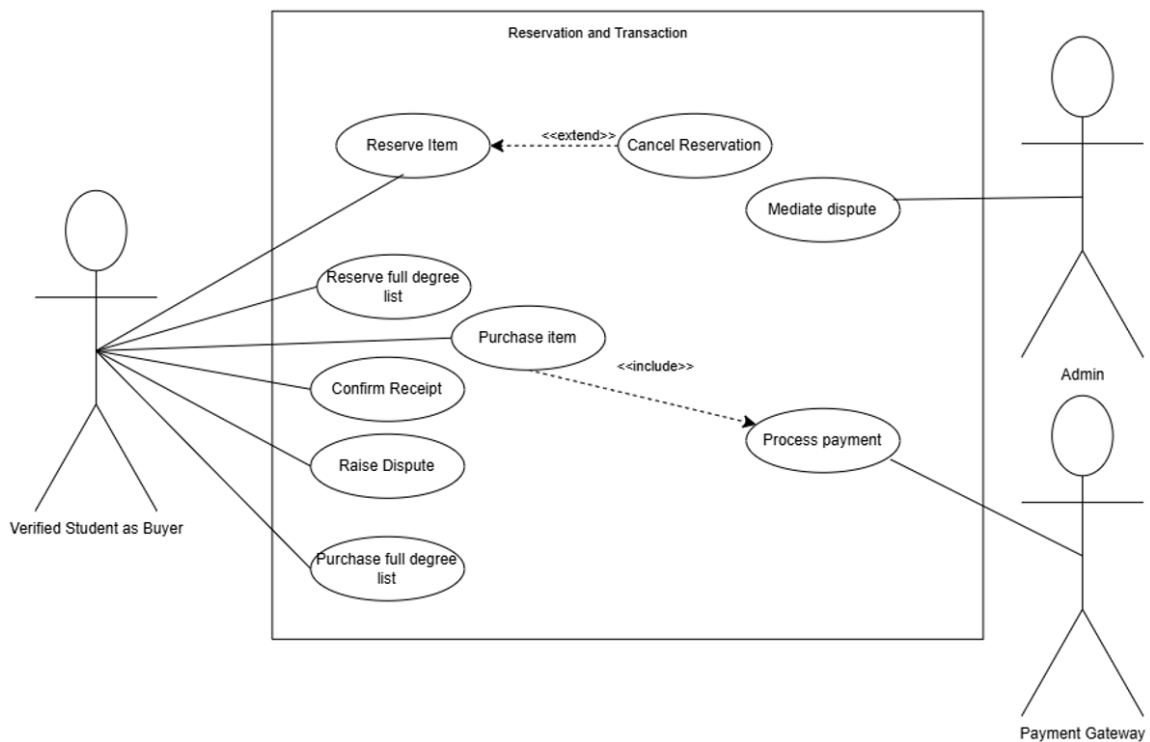
- **TUCEW** the complaint/evidence is compiled into a case record for processing by an admin.

UC4.8: Purchase full degree list

- **TUCBW** a buyer reserves a full degree list from a seller.
- **TUCEW** a 7-day timer is started for the reservation, and a chat thread is also open where the buyer can communicate with the seller.

UC4.9: Mediate dispute

- **TUCBW** an admin opens a reported/flagged dispute case.
- **TUCEW** and admin reaches a decision, applying a seller strike, listing removal, or account action.



UC5: Admin and Dispute Resolution

UC5.1: View pending verifications

- **TUCBW** the admin accesses the verification queue containing students' Proof of Registration.
- **TUCEW** an admin selects a specific student verification case to inspect.

UC5.2: Approve/reject proof of registration

- **TUCBW** the admin evaluates an escalated Proof of Registration document.
- **TUCEW** the admin makes a decision, and the user is notified via email.

UC5.3: View AI failed listings

- **TUCBW** the admin opens the moderation queue for high-risk or flagged marketplace listings.
- **TUCEW** the admin loads the details, images, seller history, and AI risk reasons for a flagged listing.

UC5.4: Manually approve/reject materials

- **TUCBW** the admin reviews a flagged/high-risk listing.
- **TUCEW** the admin selects Approve (listing goes live), Reject (listing removed, seller notified), or Request changes (listing sent back to draft).

UC5.5: View dispute cases

- **TUCBW** the admin opens the dispute management dashboard.
- **TUCEW** the system displays all the compiled cases.

UC5.6: Review evidence

- **TUCBW** the admin opens a specific dispute file to examine available evidence.
- **TUCEW** the admin finishes evaluating all provided evidence and proceeds to decision-making.

UC5.7: Mediate and decide outcome

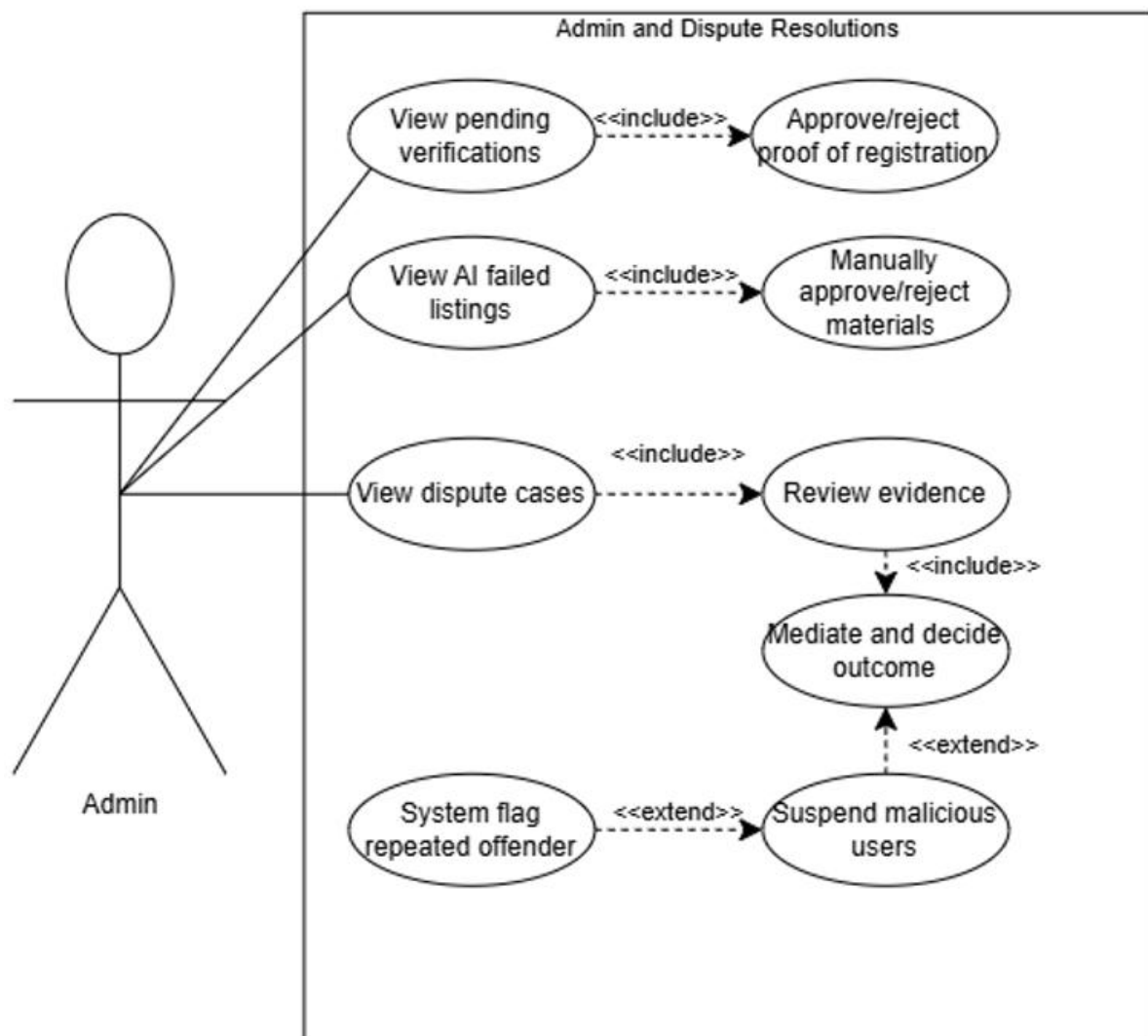
- **TUCBW** the admin weighs the evidence and user historical patterns against platform policies.
- **TUCEW** the admin issues a final verdict (e.g. strike assigned, listing removed, or dispute dismissed).

UC5.8: System flag repeated offender

- **TUCBW** the system records cumulative negative signals exceeding thresholds limits.
- **TUCEW** the system automatically logs severe risk flag against the account or triggers escalation.

UC5.9: Suspend malicious users

- **TUCBW** the repeated offender threshold is crossed, or an Admin manually triggers suspension after dispute mediation.
- **TUCEW** the malicious account is restricted/suspended from the platform, and trust (reputation score) are updated.



UC6: Communication and Logistics

UC6.1: Open chat thread

- **TUCBW** a buyer presses “Reserve” on the listing’s card of the listing’s details page; a seller acknowledges and click messages on their side bar.
- **TUCEW** a direct, private messaging channel between the buyer and seller is established.

UC6.2: Send message

- **TUCBW** a user types and sends text/coordination details in the chat thread.
- **TUCEW** the message is delivered to the recipient and rendered in the chat view.

UC6.3: Get push notification

- **TUCBW** an event occurs, 15-minute meetup check-in reminder, listing status change, or new chat message.
- **TUCEW** the system delivers a push notification to the user’s device.

UC6.4: Enter seller’s generated pin

- **TUCBW** the buyer completes payment via PayFast, seller receives a system-generated PIN.
- **TUCEW** the buyer enters and submits the PIN, confirming successful handover and completing the transaction.

UC6.5: Add Review

- **TUCBW** a transaction is closed via PIN verification, and it appears on the buyer’s My Orders page and the seller’s My Sales page.
- **TUCEW** the buyer/seller submits a start rating and feedback for the transaction.

UC6.6: Provide Location

- **TUCBW** the buyer/seller taps “I’m here” within the 15-minute pre-meetup to 30-minute post-meetup window.
- **TUCEW** the app requests GPS coordinates via Leaflet, verifies proximity, and logs the verified check-in timestamp.

